

AUTO-LOAN SECURITISATION PORTFOLIO ANALYSIS

Power BI Data Analytics Project – Final Report

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Executive Summary

This project develops a Power BI-based analytical framework for evaluating an auto-loan securitisation portfolio. The solution integrates portfolio performance analysis, delinquency monitoring, vintage analysis, IFRS 9 Expected Credit Loss (ECL) assessment, stress testing, waterfall and tranche analysis, investor reporting, and rating-agency data tape preparation.

The project follows a structured data-modeling approach using a star-schema architecture and DAX-based analytical measures. The Power BI model provides portfolio-level KPIs, delinquency trends, loss and recovery analysis, prepayment analysis, credit-risk indicators, and stress-scenario analysis.

Validation was performed through Excel cross-checks and DAX Studio performance testing. The ECL calculation was cross-checked against an Excel calculation, while waterfall allocations were independently validated using manual Excel calculations. Key DAX queries were also tested for performance against the project's target of less than 100 milliseconds.

The final solution is designed to support portfolio monitoring, risk analysis, investor reporting, and data-driven decision-making throughout the securitisation lifecycle.

Project Objective

The primary objective of this project is to develop an interactive Power BI analytical solution for monitoring and evaluating an auto-loan securitisation portfolio.

The project aims to:

- Analyse overall loan portfolio performance and outstanding balances.
- Monitor delinquency and Days Past Due (DPD) behaviour.
- Analyse portfolio performance across different loan vintages.
- Calculate and analyse IFRS 9 Expected Credit Loss (ECL).
- Evaluate portfolio behaviour under different stress scenarios.
- Analyse waterfall allocation and tranche-level credit enhancement.
- Provide investor-focused reporting and portfolio summaries.
- Prepare a loan-level data tape suitable for rating-agency reporting requirements.
- Validate important calculations using Excel and DAX Studio.
- Implement role-based access control (RLS) within the Power BI model.

Securitisation & Auto Loan Portfolio Overview

Auto-loan securitisation involves pooling individual auto loans and analysing the resulting portfolio based on its outstanding balance, cash flows, delinquency behaviour, losses, recoveries, prepayments, and credit-risk characteristics.

The project analyses an auto-loan portfolio using loan-level and monthly portfolio data. The Power BI model brings together information related to loans, borrowers, vehicle characteristics, delinquency snapshots, monthly losses, and vintage performance.

The analysis focuses on key portfolio indicators such as total loans, original balance, current outstanding balance, collections, overdue amounts, delinquency levels, gross losses, recoveries, prepayments, and net losses.

The portfolio is further analysed by vehicle type, loan characteristics, delinquency buckets, reporting periods, and loan vintages. These analyses provide the foundation for subsequent credit-risk, ECL, stress-testing, waterfall, and investor-reporting sections of the project.

Data & Star Schema

The Power BI solution uses a structured data model to support portfolio analysis and reporting. The model connects loan-level information with borrower, vehicle, date, delinquency, loss, and vintage-related data.

A star-schema-oriented structure is used to improve analytical performance and simplify DAX calculations. Dimension tables provide descriptive attributes, while transactional and snapshot data provide the measures required for portfolio analysis.

The model includes dedicated dimensions for loans, borrowers, vehicles, and dates. The date dimension supports time-based analysis such as monthly trends, rolling-period calculations, and year-to-date analysis.

Relationships between the tables allow portfolio-level measures to be analysed across different dimensions while maintaining a consistent analytical structure.

Portfolio Performance Analysis

The Power BI dashboard provides a consolidated view of the auto-loan portfolio through key performance indicators and analytical visuals. The main portfolio measures include Total Loans, Original Balance, Total Outstanding, Collections, Net Loss, and Amount Overdue.

Portfolio performance is analysed across reporting periods and vehicle types to identify changes in outstanding balances and portfolio composition. Monthly trend analysis is used to monitor movements in portfolio balances and net losses.

The dashboard also provides a comparison of current and repossessed portfolio positions and analyses overdue balances across different DPD buckets. These indicators provide a consolidated view of portfolio quality and performance.

DPD & Delinquency Analysis

Delinquency analysis is used to monitor the repayment performance and credit quality of the auto-loan portfolio. The analysis categorises loans according to their Days Past Due (DPD) buckets and measures the corresponding number of loans and outstanding balances.

The dashboard provides DPD metrics including 30+ DPD, delinquency rates, overdue balances, and rolling-period delinquency measures. DPD trends are analysed over time to identify changes in portfolio delinquency behaviour.

The Investor Reporting section also provides a delinquency report showing the number of loans, delinquent balance, and percentage of the pool represented by each DPD bucket. These measures support ongoing portfolio monitoring and reporting.

Vintage Analysis

Vintage analysis evaluates portfolio performance by grouping loans according to their origination or vintage period. This approach helps compare the cumulative performance of different loan cohorts over time.

The Power BI model uses vintage-level data to analyse cumulative gross losses, recoveries, and net losses. Cumulative net loss rates are compared across vintages to identify differences in portfolio performance.

The dashboard also includes analysis of average DPD age, providing an additional view of delinquency behaviour across the portfolio.

Vintage analysis supports the identification of performance patterns across different origination cohorts and provides an additional perspective for portfolio risk monitoring.

IFRS 9 Expected Credit Loss (ECL)

The project incorporates an IFRS 9 Expected Credit Loss (ECL) analysis to estimate credit losses associated with the auto-loan portfolio. The analysis considers Exposure at Default (EAD), Probability of Default (PD), and Loss Given Default (LGD).

The Power BI model calculates Stage 1 12-month ECL and lifetime ECL based on the available portfolio data and IFRS 9 staging information. EAD is also analysed across IFRS 9 stages to provide visibility into the distribution of portfolio exposure.

The ECL calculation was independently cross-checked using an Excel validation workbook. The DAX result and Excel calculation were compared to verify the calculation and quantify the variance.

Stress Testing

Stress testing is used to evaluate how changes in key portfolio assumptions affect expected credit losses. The Power BI solution provides interactive stress scenarios using What-If parameters.

The model includes stress parameters for default rate, recovery rate haircut, prepayment speed, and interest-rate shock. Scenario analysis allows the impact of changing assumptions to be observed through adjusted ECL and related risk indicators.

The dashboard presents Base, Moderate, Severe, and Crisis ECL scenarios. This provides a structured framework for analysing potential changes in credit-risk exposure under different assumptions.

The stress-testing measures were also tested in DAX Studio to assess calculation performance.

Waterfall & Tranche Analysis

The project includes a waterfall analysis to evaluate how portfolio collections are allocated across different securitisation tranches.

The model calculates Senior Tranche Allocation, Mezzanine Tranche Allocation, and Equity Residual Allocation based on available collections and the defined tranche allocation structure. Credit enhancement and Overcollateralisation (OC) ratios are also calculated to monitor tranche protection.

Trigger-status measures are used to identify whether defined OC thresholds are met. The waterfall allocation was independently validated in Excel by comparing DAX-generated allocations with manual calculations.

The validation confirmed that the DAX allocation and manual Excel allocation produced matching results for the tested scenario.

Investor Reporting & Data Type

The Investor Reporting dashboard provides structured reporting across key portfolio areas including Pool Summary, Delinquency Report, Loss & Recovery Report, Prepayment Report, Waterfall Report, and Trigger Status.

The Pool Summary provides an overview of loan count, balances, collections, and net losses. The Delinquency Report presents loan counts and balances by DPD bucket. Loss and recovery analysis provides cumulative loss and recovery information across portfolio vintages.

The Prepayment Report analyses prepayment amounts and rates over reporting periods, while the Waterfall Report provides tranche allocation and trigger information.

A loan-level data tape was also prepared for rating-agency reporting purposes. The data tape contains available loan, borrower, vehicle, delinquency, performance, and credit-risk fields and was exported to CSV format.

RLS, Governance & Validation

Role-Based Security (RLS) was implemented in the Power BI model using five defined roles: Auditor, Portfolio Manager, Risk Analyst, Senior Management, and Servicer. Region-based rules were configured for these roles.

Calculation and model validation were performed using Excel and DAX Studio. The ECL calculation was cross-checked against an Excel calculation, and waterfall allocations were compared with manual Excel calculations.

DAX Studio was also used to evaluate query performance. The tested analytical queries remained below the project target of 100 milliseconds.

A DAX Measure Dictionary was prepared to document the measures used in the Power BI model, and a DAX Performance report was maintained as part of the validation workbook.

Validation Note: RLS role creation was completed, while full filter-propagation validation across all related tables was not completed during the project.

2008 Global Financial Crisis Case Study

The 2008 Global Financial Crisis demonstrated how weaknesses in lending standards, securitisation structures, credit-risk assessment, and excessive leverage can contribute to significant financial-system stress.

In the context of securitisation, the crisis highlighted the importance of monitoring underlying loan quality, delinquency, defaults, losses, recoveries, and credit enhancement. It also demonstrated the importance of transparency and reliable reporting for investors and other stakeholders.

The lessons from the crisis are relevant to this project because the Power BI solution focuses on portfolio-level monitoring, delinquency analysis, loss and recovery analysis, ECL assessment, stress testing, and tranche-level indicators.

These analytical components provide a framework for identifying changes in portfolio risk and evaluating potential impacts under stressed conditions.

IL&FS Case Study

The Infrastructure Leasing & Financial Services (IL&FS) crisis is an important Indian case study for understanding liquidity risk, asset-liability mismatches, interconnected financial exposures, and the importance of effective risk monitoring.

The case highlights the importance of monitoring cash flows, repayment obligations, credit quality, concentration risk, and liquidity conditions. It also demonstrates how deterioration in financial conditions can affect multiple stakeholders across a financial network.

The relevance to this project lies in the need for continuous portfolio monitoring and early identification of risk indicators. Measures such as delinquency, losses, recoveries, exposure concentration, stress testing, and cash-flow-related analysis can support better risk monitoring and reporting.

The case study therefore reinforces the importance of timely data, transparent reporting, robust risk controls, and scenario-based analysis in financial portfolios.

Conclusion

The project developed a comprehensive Power BI solution for analysing an auto-loan securitisation portfolio. The solution combines portfolio performance monitoring, DPD analysis, vintage analysis, IFRS 9 ECL, stress testing, waterfall and tranche analysis, investor reporting, and loan-level data tape preparation.

The analytical model was supported by DAX measures and a structured data model. Key calculations were validated through Excel cross-checks, while DAX Studio was used to evaluate query performance against the project's performance target.

The completed solution provides a consolidated analytical environment for monitoring portfolio behaviour, credit-risk indicators, losses, recoveries, and stress scenarios. The project also demonstrates the importance of data validation, reporting transparency, security controls, and performance optimisation in financial analytics.

Appendix – DAX Measure & Validation

The Power BI model contains a comprehensive set of DAX measures supporting portfolio analysis, delinquency monitoring, time intelligence, vintage analysis, ECL, stress testing, concentration analysis, waterfall allocation, investor reporting, and other analytical requirements.

A DAX Measure Dictionary was created to document the measures used in the model. DAX Studio was used to test representative queries for calculation performance, with the tested queries remaining below the project target of 100 milliseconds.

Excel validation workbooks were prepared for key financial calculations, including IFRS 9 ECL and waterfall allocation. These validation files provide an independent comparison between Power BI/DAX calculations and manual or Excel-based calculations.

References & Data Sources

1. Project 1 Internship Roadmap and Requirements Document
2. Auto-Loan Securitisation Portfolio Dataset
3. Power BI Data Model and DAX Measures
4. Excel Validation Workbook – ECL Crosscheck and Waterfall Validation
5. DAX Studio Performance Testing Results

Regulatory References

Reserve Bank of India – Master Direction on Securitisation of Standard Assets, 2021;
Securities and Exchange Board of India – Regulations on Issue and Listing of Securitised Debt Instrument and Security Receipts.